

DATA MINING – LAB

Section (A)

ASSIGNMENT # 1

Group members :

Fatima Sohail	231980004
Namra Noor	231980046

Titanic Survival Data – Data Cleaning & Preparation Report

1. Introduction

The objective of this task was to perform data cleaning and preparation on the Titanic Survival Dataset. The focus was on handling missing values, engineering useful features, and encoding categorical variables to prepare the dataset for further analysis or modeling.

2. Dataset Overview

The dataset contains **891 passengers** with the following key columns:

- PassengerId – Unique ID for each passenger
- Survived – Target variable (0 = Not survived, 1 = Survived)
- Pclass – Passenger class (1 = 1st, 2 = 2nd, 3 = 3rd)
- Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked

Initially :

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   PassengerId      891 non-null    int64
1   Survived         891 non-null    int64
2   Pclass          891 non-null    int64
3   Name             891 non-null    object
4   Sex              891 non-null    object
5   Age              714 non-null    float64
6   SibSp            891 non-null    int64
7   Parch            891 non-null    int64
8   Ticket           891 non-null    object
9   Fare             891 non-null    float64
10  Cabin            204 non-null    object
11  Embarked         889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

```
[12]: PassengerId      0
      Survived       0
      Pclass         0
      Name           0
      Sex            0
      Age            177
      SibSp          0
      Parch          0
      Ticket         0
      Fare           0
      Cabin          687
      Embarked       2
      dtype: int64
```

3. Preprocessing Steps

3.1 Handling Missing Values

- Age: Filled missing values with the median age, as age distribution can have outliers.
- Embarked: Filled missing values with the mode (most frequent port).
- Fare: Filled missing values with the median fare.

- Cabin: Instead of using the cabin directly, created a new column CabinKnown (1 if cabin info exists, 0 if missing) and dropped the original Cabin column.

3.2 Feature Engineering

Several new features were created to improve dataset quality:

1. Title Extraction – Extracted titles from the Name column (e.g., Mr, Mrs, Miss) and grouped rare titles into a single Rare category.
2. Family Size – Calculated as:

```
df['FamilySize'] = df['SibSp'] + df['Parch'] + 1
```

This counts the total number of family members on board including the passenger themselves.

3. Alone Indicator – Created Alone column: 1 if the passenger is traveling alone (FamilySize == 1), 0 otherwise.
4. Ticket Group Size – Calculated the number of passengers sharing the same ticket using TicketGroup.

3.3 Encoding Categorical Variables

- Converted Sex, Embarked, and Title into dummy/indicator variables using `pd.get_dummies()`.
- Dropped the first category in each case to avoid multicollinearity.

3.4 Dropping Unnecessary Columns

- Removed Name and Ticket columns, as they are not useful for analysis after feature engineering.

4. Final Cleaned & Prepared Dataset

The final cleaned dataset contains **20 columns**:

- Numeric columns: PassengerId, Survived, Pclass, Age, SibSp, Parch, Fare, CabinKnown, FamilySize, Alone, TicketGroup

- Encoded categorical columns (bool type): Sex_male, Embarked_Q, Embarked_S, Title_Miss, Title_Mr, Title_Mrs, Title_Rare

Sample output:

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare	CabinKnown	\
0	1	0	3	22.0	1	0	7.2500	0	
1	2	1	1	38.0	1	0	71.2833	1	
2	3	1	3	26.0	0	0	7.9250	0	
3	4	1	1	35.0	1	0	53.1000	1	
4	5	0	3	35.0	0	0	8.0500	0	

	FamilySize	Alone	TicketGroup	Sex_male	Embarked_Q	Embarked_S	\
0	2	0	1	True	False	True	
1	2	0	1	False	False	False	
2	1	1	1	False	False	True	
3	2	0	2	False	False	True	
4	1	1	1	True	False	True	

	Title_Miss	Title_Mr	Title_Mrs	Title_Rare
0	False	True	False	False
1	False	False	True	False
2	True	False	False	False
3	False	False	True	False
4	False	True	False	False

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 891 entries, 0 to 890

Data columns (total 20 columns):

#	Column	Non-Null Count	Dtype
0	PassengerId	891 non-null	int64
1	Survived	891 non-null	int64
2	Pclass	891 non-null	int64
3	Name	891 non-null	object
4	Age	891 non-null	float64
5	SibSp	891 non-null	int64
6	Parch	891 non-null	int64
7	Ticket	891 non-null	object
8	Fare	891 non-null	float64

```

9   CabinKnown      891 non-null    int64
10  FamilySize      891 non-null    int64
11  Alone           891 non-null    int64
12  TicketGroup     891 non-null    int64
13  Sex_male        891 non-null    bool
14  Embarked_Q      891 non-null    bool
15  Embarked_S      891 non-null    bool
16  Title_Miss      891 non-null    bool
17  Title_Mr        891 non-null    bool
18  Title_Mrs       891 non-null    bool
19  Title_Rare      891 non-null    bool
dtypes: bool(7), float64(2), int64(9), object(2)
memory usage: 96.7+ KB

```

```

[17]: PassengerId      0
      Survived        0
      Pclass          0
      Name            0
      Age             0
      SibSp           0
      Parch           0
      Ticket          0
      Fare            0
      CabinKnown      0
      FamilySize      0
      Alone           0
      TicketGroup     0
      Sex_male        0
      Embarked_Q      0
      Embarked_S      0
      Title_Miss      0
      Title_Mr        0
      Title_Mrs       0
      Title_Rare      0
      dtype: int64

```

- All missing values have been handled.
- Dataset is now ready for analysis or machine learning tasks.

5. Summary

- Missing values were handled using median and mode strategies.
- Feature engineering improved dataset quality and added meaningful information like FamilySize, Alone, and Title.
- Categorical variables were encoded for analysis.
- Final dataset is cleaned, fully prepared, and contains no missing values.

The preprocessing ensures the dataset is ready for predictive modeling or further exploratory analysis.